

MERI NUKALA

Embedded & IoT Enthusiast

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PROFESSIONAL SUMMARY

Enthusiastic B.Tech Electronics and Communication Engineering student with hands-on experience in Embedded Systems, Arduino, ESP32, ESP8266, and sensor-based robotics. Skilled in C, Embedded C, and Python with knowledge of microcontrollers, GPIO, ADC, PWM, timers, interrupts, communication protocols, and sensor interfacing. Seeking Embedded Systems/IoT opportunities to apply real-time problem-solving, hardware-software integration, testing, and debugging skills.

EDUCATION

ISTS Women's Engineering College, JNTUK	2023–2027
B.Tech in Electronics & Communication Engineering	Rajahmundry, Andhra Pradesh
Vijaya Kranthi Junior College, Challapalli	2021–2023
Intermediate (MPC)	92.1%
Nagarjuna Public School, Krishna District	2020–2021
SSC	98.83%

TECHNICAL SKILLS

Programming: C, Embedded C, Python, Java

Embedded Systems: Microcontrollers, GPIO, ADC, DAC, PWM, Timers, Counters, Interrupts, Watchdog Timer, Sensor Interfacing, Actuator Interfacing

Boards: Arduino, ESP32, ESP8266, 8051

Communication Protocols: UART, SPI, I2C, Serial Communication

Sensors & Hardware: PIR, Ultrasonic, IR, Relay, LED, Buzzer, Potentiometer, Basic PCB Concepts

IoT: IoT Fundamentals, Wi-Fi Communication, Sensor Data Acquisition, IoT Applications

Tools: Arduino IDE, Keil μ Vision, Proteus, Tinkercad, MATLAB, VS Code, Git, GitHub

INTERNSHIP

Embedded and IoT Analyst **July 2026 – August 2026**

- Gained hands-on experience in embedded systems, IoT concepts, sensor interfacing, microcontrollers, and hardware-software integration.
- Worked with embedded development, testing, debugging, and real-time sensor-based applications using Arduino/IoT technologies.

PROJECTS

Ultrasonic Sensor-Based Speed Control Robotic Car | Arduino, Embedded C  GitHub

- Designed and developed an Arduino-based robotic car that uses ultrasonic distance sensing to detect obstacles and adjust motor speed in real time.
- Implemented GPIO, ultrasonic sensor interfacing, distance measurement, and PWM-based motor control with Embedded C for responsive robotic operation.

Smart Dustbin Using Arduino  GitHub

- Designed an embedded automation system using Arduino, ultrasonic sensor, and servo motor for touchless waste disposal and real-time object detection.
- Implemented GPIO programming, ultrasonic sensor interfacing, distance measurement, threshold-based control, servo motor control, and hardware testing.

Soil Moisture Testing and Monitoring System

- Designed an Arduino-based soil moisture monitoring system using an analog soil moisture sensor for real-time soil condition measurement.
- Implemented ADC-based sensor interfacing, threshold detection, and real-time monitoring for smart agriculture and IoT applications.

CERTIFICATIONS

NPTEL: The Joy of Computing Using Python – 2025

Infosys Springboard: Basics of Python – 2025; Introduction to Python – 2025

Infosys Springboard: Machine Learning with Python – 2025

ServiceNow: Introduction to Generative AI

ADDITIONAL SKILLS

Problem Solving | Debugging | Hardware Testing | Troubleshooting | Circuit Analysis | Datasheet Reading | Embedded Systems Testing | Technical Documentation | Teamwork | Communication | Quick Learning